

EXPERIMENT 2

Aim

Convert the ER diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Relational Schema

Customer (CustomerID PK, Name, Email UNIQUE, PhoneNo, DOB)	Address (AddressID PK, CustomerID FK, Pincode, State, City)
Seller (SellerID PK, SellerName, PhoneNo UNIQUE)	Category (CategoryID PK, CategoryName UNIQUE, Description)
Product (ProductID PK, CategoryID FK, Price, Image)	Electronic (ProductID PK, FK)
Fashion (ProductID PK, FK)	Books (ProductID PK, FK, SellerID FK)
Orders (OrderID PK, CustomerID FK, OrderDate, OrderStatus)	Order_Item (OrderID PK, FK, item_seq_no PK, ProductID FK, Quantity, Unit_Price)
Payment (PaymentID PK, OrderID FK, UNIQUE, PaymentDate, Amount, PaymentMethod, Status)	Delivery (DeliveryID PK, OrderID FK, UNIQUE, CourierName, Status, DeliveryDate)

Source Code

1. Database Creation

```
CREATE DATABASE EcommerceDB;  
USE EcommerceDB;
```

2. Table Creation

```
CREATE TABLE Customer (  
    CustomerID INT PRIMARY KEY AUTO_INCREMENT,  
    Name       VARCHAR(100) NOT NULL,  
    Email      VARCHAR(100) NOT NULL UNIQUE,  
    PhoneNo    VARCHAR(15)  NOT NULL,  
    DOB        DATE NOT NULL  
);  
  
CREATE TABLE Address (  
    AddressID INT PRIMARY KEY AUTO_INCREMENT,  
    CustomerID INT NOT NULL,  
    Pincode    VARCHAR(10) NOT NULL,  
    State      VARCHAR(50) NOT NULL,  
    City       VARCHAR(50) NOT NULL,  
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)  
        ON DELETE CASCADE  
);  
  
CREATE TABLE Category (  
    CategoryID INT PRIMARY KEY AUTO_INCREMENT,  
    CategoryName VARCHAR(100) NOT NULL UNIQUE,  
    Description  VARCHAR(255)  
);  
  
CREATE TABLE Seller (  
    SellerID INT PRIMARY KEY AUTO_INCREMENT,  
    SellerName VARCHAR(100) NOT NULL,  
    PhoneNo    VARCHAR(15)  NOT NULL UNIQUE  
);  
  
CREATE TABLE Product (  
    ProductID INT PRIMARY KEY AUTO_INCREMENT,  
    CategoryID INT NOT NULL,  
    Price      DECIMAL(10,2) NOT NULL,  
    Image      VARCHAR(255),  
    FOREIGN KEY (CategoryID) REFERENCES Category(CategoryID)  
        ON DELETE CASCADE  
);  
  
CREATE TABLE Electronic (  
    ProductID INT PRIMARY KEY,  
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)  
        ON DELETE CASCADE  
);  
  
CREATE TABLE Fashion (  
    ProductID INT PRIMARY KEY,  
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
```

```

        ON DELETE CASCADE
    );

CREATE TABLE Books (
    ProductID INT PRIMARY KEY,
    SellerID INT,
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
        ON DELETE CASCADE,
    FOREIGN KEY (SellerID) REFERENCES Seller(SellerID)
        ON DELETE SET NULL
);

CREATE TABLE Orders (
    OrderID INT PRIMARY KEY AUTO_INCREMENT,
    CustomerID INT NOT NULL,
    OrderDate DATE NOT NULL,
    OrderStatus VARCHAR(30) NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);

CREATE TABLE OrderItem (
    OrderID INT NOT NULL,
    item_seq_no INT NOT NULL,
    ProductID INT NOT NULL,
    Quantity INT NOT NULL,
    Unit_Price DECIMAL(10,2) NOT NULL,
    PRIMARY KEY (OrderID, item_seq_no),
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
        ON DELETE CASCADE,
    FOREIGN KEY (ProductID) REFERENCES Product(ProductID)
        ON DELETE CASCADE
);

CREATE TABLE Payment (
    PaymentID INT PRIMARY KEY AUTO_INCREMENT,
    OrderID INT NOT NULL UNIQUE,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentMethod VARCHAR(30) NOT NULL,
    Status VARCHAR(30) NOT NULL,
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
        ON DELETE CASCADE
);

CREATE TABLE Delivery (
    DeliveryID INT PRIMARY KEY AUTO_INCREMENT,
    OrderID INT NOT NULL UNIQUE,
    CourierName VARCHAR(100) NOT NULL,
    Status VARCHAR(30) NOT NULL,
    DeliveryDate DATE,
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
        ON DELETE CASCADE
);

```

Inserting Sample Data

```
INSERT INTO Customer (Name, Email, PhoneNo, DOB) VALUES
('Sahil', 'sahil@gmail.com', '9839105025', '2005-05-15'),
('Arjun', 'arjun@gmail.com', '9336677989', '2004-08-20');
```

```
INSERT INTO Category (CategoryName, Description) VALUES
('Electronics', 'Electronic Products'),
('Fashion', 'Fashion Products'),
('Books', 'Books');
```

```
INSERT INTO Seller (SellerName, PhoneNo) VALUES
('Tech Store', '9000000001'),
('Book Store', '9000000002');
```

```
INSERT INTO Product (CategoryID, Price, Image) VALUES
(1, 50000, 'Laptop.jpg'),
(2, 999, 'Shirt.png'),
(3, 599, 'Book.jpg');
```

```
INSERT INTO Electronic VALUES (1);
INSERT INTO Fashion VALUES (2);
INSERT INTO Books VALUES (3, 2);
```

```
INSERT INTO Address (CustomerID, Pincode, State, City) VALUES
(1, '208001', 'Uttar Pradesh', 'Kanpur'),
(2, '110001', 'Delhi', 'Delhi');
```

```
INSERT INTO Orders (CustomerID, OrderDate, OrderStatus) VALUES
(1, '2026-08-19', 'Placed'),
(2, '2026-08-19', 'Shipped');
```

```
INSERT INTO OrderItem VALUES
(1, 1, 1, 1, 50000),
(2, 1, 2, 2, 999);
```

```
INSERT INTO Payment (OrderID, PaymentDate, Amount, PaymentMethod, Status) VALUES
(1, '2026-08-19', 50000, 'UPI', 'Success'),
(2, '2026-08-19', 1998, 'Card', 'Success');
```

```
INSERT INTO Delivery (OrderID, CourierName, Status, DeliveryDate) VALUES
(1, 'Delhivery', 'Delivered', '2026-08-20'),
(2, 'DTDC', 'Pending', NULL);
```

Referential Integrity Violations

Case 1: Foreign Key Violation

```
INSERT INTO Orders (CustomerID, OrderDate, OrderStatus) VALUES  
(999, '2026-08-19', 'Placed');
```

Output:

```
ERROR 1452 (23000): Cannot add or update a child row:  
a foreign key constraint fails
```

Case 2: Unique Constraint Violation

```
INSERT INTO Customer (Name, Email, PhoneNo, DOB) VALUES  
('Test', 'sahil@gmail.com', '9999999999', '2025-01-01');
```

Output:

```
ERROR 1062 (23000): Duplicate entry 'sahil@gmail.com'  
for key 'Customer.Email'
```

Verification Queries

```
SELECT * FROM Customer;  
SELECT * FROM Product;  
SELECT * FROM Orders;  
SELECT * FROM OrderItem;
```

ON DELETE SET NULL

```
DELETE FROM Seller WHERE SellerID = 2;  
SELECT * FROM Books;
```

ON DELETE CASCADE

```
DELETE FROM Customer WHERE CustomerID = 2;  
SELECT * FROM Orders WHERE CustomerID = 2;
```

Output

Table: Customer

CustomerID	Name	Email	PhoneNo	DOB
1	Sahil	sahil@gmail.com	9839105025	2005-05-15
2	Arjun	arjun@gmail.com	9336677989	2004-08-20

Table: Product

ProductID	CategoryID	Price	Image
1	1	50000.00	Laptop.png
2	2	999.00	Shirt.png
3	3	599.00	Book.png

Table: Orders

OrderID	CustomerID	OrderDate	OrderStatus
1	1	2026-08-19	Placed
2	2	2026-08-19	Shipped

Table: OrderItem

OrderID	item_seq_no	ProductID	Quantity	Unit_Price
1	1	1	1	50000.00
2	1	2	2	999.00

Table: Books (after DELETE FROM Seller WHERE SellerID = 2)

ProductID	SellerID
3	NULL

Orders of CustomerID = 2 (after ON DELETE CASCADE)

Empty set

Result

The ER diagram was successfully converted into a relational schema and implemented in MySQL. All constraints — PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL — were applied and verified. Sample data was inserted, and referential integrity violations were successfully demonstrated through foreign key and unique constraint errors.